Detecting Fake News and Misinformations on Social Media Platform

Albert Wang
albert@vt.edu
Virginia Tech
USA

Kevin Lin
kevinlin04@vt.edu
Virginia Tech
USA

Kaitlyn Vu
kaitlynvu18@vt.edu
Virginia Tech
USA

Kelly Yap
kelly@vt.edu
Virginia Tech
USA

Qitao Yang
yqitao@vt.edu
Virginia Tech
USA

Yufei Chen
yufei0320@vt.edu
Virginia Tech
USA

1 Problem Idea

Social media is heavily integrated into our daily lives. It is so well integrated that it has become a primary source of information for users to know about the world, the news, their friends, advice, hacks, trends, and much more. This is due to the ease of accessibility and the way social media platforms allow users to stay up to date in real time. With how quickly information can spread across the world, this poses questions and concerns on whether the information that we see on our devices is credible or accurate. The problem we are investigating is detecting misleading or false information, commonly known as fake news, through social media platforms, specifically on Twitter. We believe that this problem is important as it can negatively impact the perceptions of the people, cause people to become weary and lose their trust, and bury the truth. As misinformation can be developed and spread as fast as misinformation legitimate information, we believe it is critical to propose or develop methods that could identify misinformation within social media platforms and potentially mitigate them. We believe that doing so with this topic could ensure credibility and reliability within posts and content along with helping users gain or regain that trust in getting their news fro social media platforms. This topic is related to social media analysis, as it analyzes patterns within the message or content of the post and how users interact with that message or content. Specifically, we are planning to look at linguistic patterns within the messages or post, potentially finding the source of the misinformation, and examining different strategies and techniques to identify and filter misinformation.

2 Motivation

The rapid spread of misinformation online has become an increasing worry as social media usage rises. Platforms such as Facebook and Twitter are designed to support immediate interaction and effortless content sharing across large networks. While this design increases user engagement, it also enables misinformation to spread widely before it can be verified.

Highlighting the extent of this issue is a study conducted by researchers at Penn State University where they analyzed millions of posts containing links shared on Facebook between 2017 and 2020. The study found that approximately 75% of shares were made without users clicking the link first [1]. This behavior suggests that users tend to share information without fact checking or evaluating its credibility.

This issue is especially concerning given that social media has become a primary source of news and information for many individuals, and the consequences of misinformation spread into various domains. In political contexts, misleading content has been shown to influence public perception during elections and major events [2]. In healthcare, misinformation about vaccines and disease treatments has affected personal health decisions and trust in science [3].

On social media platforms, misinformation appears in short-form posts rather than full-length articles. These posts often rely on concise and striking language in order to increase its chances at being liked and shared. As a result, typical approaches that focus on analyzing full articles or long-form content may not be sufficient for detecting misinformation. This creates a need for methods that can operate directly at the level of social media posts. This project is motivated by the goal of developing a model capable of detecting fake news and misinformation within social media content, providing a more appropriate approach for the way information is created and shared on these platforms.

3 Dataset

The dataset that we plan to use is the CIC truth seeker dataset. This dataset is from the Canadian Institute for Cybersecurity, and it aims to be the largest ground fake news analysis dataset in relation to social media. It contains more than 180,000 labeled tweets, all of which were verified using a three factor verification method with 456 Amazon Mechanical Turkers for labeling each tweet.

This dataset if appropriate for our research because it not only provides the actual text, but it also includes a number of lexical features like number of present verbs, past verbs, adjectives, as well as text features like the number of unique, complex words. It also introduces bot, credibility, and influence scores to better help understand patterns of Twitter users. All of this combined gives us a wide range of information to create a robust system.

3.1 Dataset Examples

Key variables include:

- **author**: Source of the original statement
- **statement**: News headline or claim being evaluated
- **tweet**: A Twitter post related to the statement
- **BinaryNumTarget**: Ground-truth label (1 = True, 0 = False)
- **manual_keywords**: Keywords used to retrieve related tweets

- **3_label_majority_answer / 5_label_majority_answer**: Crowd-sourced agreement labels

Example data instances:

- **Example 1**
  - **BinaryNumTarget**: 1 (True)
  - **statement**: Trump, federal minimum wage
  - **tweet**: @MichaelTuckness @LeeStls33 @Interloper_X @NicoleMarie1081 @restore_rescue @DonaldJTrumpJr Biden released his 2021 tax returns after Trump refused to release his for his entire 4-year term.
    Biden increased the minimum wage for thousands of federal contractors to $15/hr.
    Biden directed $2.5 billion to address America's mental illness and addiction crisis.
  - **3_label_majority_answer**: Mostly Agree
  - **5_label_majority_answer**: Agree
- **Example 2**
  - **BinaryNumTarget**: 0 (False)
  - **author**: Tom Kertscher
  - **statement**: Joe Biden's great-grandfather Joseph J. Biden was a slave owner and fought for the Confederacy.
  - **tweet**: So Kamala Harris wants to be Biden's VP, yet she said she believed his accusers—what a hypocrite. She also supports BLM, yet forgets to mention that her great-grandfather was one of the largest slave owners in Jamaica 80 years after the US banned slavery.
  - **manual_keywords**: Biden, great grandfather, slave owner
  - **3_label_majority_answer**: Agree
  - **5_label_majority_answer**: Agree
- **Example 3**
  - **BinaryNumTarget**: 1 (True)
  - **author**: Ciara O'Rourke
  - **statement**: Says Spectrum will provide free internet to students during coronavirus school closures.
  - **tweet**: Spectrum giving two months of free internet for college students—that is $100 saved, I need that.
  - **manual_keywords**: Spectrum, free internet, students
  - **3_label_majority_answer**: No Majority
  - **5_label_majority_answer**: Agree

4 Approach

For this project, we assess various types of text representations as well as various classification methods in order to detect fake news on social media.

First, we perform text processing on the tweets using text cleaning operations such as lower casing, removal of punctuation marks, and tokenization.

Then, we try out various text representation techniques including TF-IDF, which is commonly used to extract word frequency features from a document. Other methods include encoding semantics through Word2Vec and GloVe, along with pre-trained BERT representation for contextual semantics.

Table 1: Experimental Setup

Exp.	Representation	Classifier
1	TF-IDF	Logistic Regression
2	TF-IDF	SVM
3	Word2Vec	Random Forest
4	GloVe	Decision Tree
5	BERT (fine-tuned)	Classification Head
6	BERT (embeddings)	SVM

From Table 1, it can be observed that six different combinations of representation techniques and classifiers are tested. Simple baseline representations such as Term Frequency-Inverse Document Frequency (TF-IDF), Word2Vec, and GloVe are combined with linear and non-linear classifiers to test their effectiveness. On the other hand, BERT is used for obtaining contextualized representations using fine-tuning and feature extraction.

In order to obtain reliable results, k-fold cross-validation is applied during model training and testing. The goal here is to determine the optimal combination for detecting misinformation on short form social media platforms.

5 Metrics

To evaluate our fake news detection models, we use several standard classification metrics. Accuracy measures the overall correctness of predictions, but it may be misleading if the dataset is imbalanced.

Precision measures how many predicted fake news instances are actually fake, and recall measures how many real fake news instances are successfully detected. Since both false positives and false negatives are important in this task, we also use the F1-score to balance precision and recall.

In addition, we use a confusion matrix to analyze prediction errors and ROC-AUC for models that output probabilities to evaluate their ability to distinguish between true and false information.

6 References

[1] A. WennersHerron, "Social media users probably won't read beyond this headline, researchers say," Penn State University, Nov. 19, 2024. [Online]. Available: https://www.psu.edu/news/research/story/social-media-users-probably-wont-read-beyond-headline-researchers-say
[2] "The real impact of fake news: the rise of political misinformation—and how we can combat its influence," Columbia University School of Professional Studies, Jan. 02, 2024. [Online]. Available: https://sps.columbia.edu/news/real-impact-fake-news-rise-political-misinformation-and-how-we-can-combat-its-influence
[3] "Health Misinformation," U.S. Department of Health And Human Services, Feb. 20, 2025. https://www.hhs.gov/surgeongeneral/reports-and-publications/health-misinformation/index.html